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A SMART WRISTBAND FOR TREMOR DETECTION AND
TREATMENT SUPPORT IN PARKINSON'S DISEASE

A Senior Design Project (EENG496) Report
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Statement of Authenticity

By submitting this senior project report, we declare that:

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2. We have not knowingly violated any ethical codes of conduct and our work has taken all ethical issues into consideration.
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Abstract

Parkinson's disease is a progressive neurological condition that mainly influences motor skills, frequently appearing as uncontrollable tremors that greatly affect patients' daily routines and overall well-being. Ongoing and precise tracking of tremor features is crucial for proper diagnosis, treatment evaluation, and long-term care. Nonetheless, traditional monitoring methods are usually restricted to clinical settings, which hinders their capacity for real-time and ongoing surveillance.

This project showcases the creation and execution of a portable, wearable device for immediate tremor detection and evaluation. The suggested system combines surface electromyography for recording muscle activity with an inertial measurement unit to monitor motion dynamics. These signals are captured and handled with an embedded microcontroller, allowing for on-device evaluation without depending on external computing systems.

The signal processing framework incorporates band-pass filtering to focus on the specific tremor frequency range and is accompanied by feature extraction methods like root mean square estimation and frequency-domain analysis. A strong detection algorithm is created by merging muscle and motion signals, enabling accurate recognition of tremor events while minimizing false positives linked to voluntary movements. The system additionally measures tremor intensity using a computed tremor level indicator, offering valuable clinical information.

In addition to detection, the project examines the practicality of incorporating a non-invasive treatment method utilizing functional electrical stimulation. The suggested architecture enables a closed-loop system where identified tremor occurrences can activate regulated stimulation signals designed to reduce tremor intensity. This underscores the system's potential not just as a monitoring resource but also as a basis for upcoming assistive treatment options.

The completed prototype functions as a battery-operated wearable device, guaranteeing portability, energy efficiency, and comfort for the user. Experimental assessment shows the system's ability to conduct real-time tremor tracking, feature extraction, and classification with acceptable accuracy. The proposed solution aids in the accessible, ongoing, and smart tracking of Parkinsonian tremors, showing significant promise for incorporation into sophisticated wearable health technologies.

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Acronyms

PD	Parkinson's Disease
TreCon	Tremor Control
EMG	Electromyography
IMU	Internal Measurement Unit
UART	Universal Asynchronous Receiver-Transmitter
TX	Transmit
RX	Receive
FFT	Fast Fourier Transform
ADC	Analog to Digital Converter
RMS	Root Mean Square
GUI	Graphical User Interface
USB	Universal Serial Bus
IOT	Internet of Things
3D	Three Dimensional
MRgFUS	MR-Guided Focused Ultrasound
HiFus	High Intensity Focused Ultrasound
FES	Functional Electrical Stimulation
EMD	Electromechanical Delay
DBS	Deep Brain Stimulation

List of Symbols

V	Volt
A	Ampere
R	Resistance
Ω	Ohm
V_{ref}	Reference Voltage
I	Current
Ω	Ohm
ω	Angular Velocity
$\Delta\theta$	Change in Angular Position
Δt	Change in Time
P	Power
F_s	Sampling Frequency
n	No. of Samples
ω_n	Normalized Cut-off Frequencies
$H(s)$	Filter Transfer Function
EMG_{rect}	Rectified EMG Signal
$X(k)$	Discrete Time Fourier Transform
M	Magnitude
$f_{\text{IMU.max}}$	Frequency Associated with the Highest Spectral Amplitude
$A_{\text{tremor}}(t)$	Instantaneous Tremor Amplitude
T	Torque

$\omega_{\text{tremor}}(t)$	Instantaneous Angular Frequency
$\phi_{\text{tremor}}(t)$	Instantaneous Phase
F	Force
G	Proportional Gain
π	180° Phase Inversion
$\Phi_{\text{compensate}}$	Phase Correction for System Delays
t	Time
PW	Pulse Width
f_{stim}	Stimulation Frequency
Hz	Hertz
Z_{skin}	Skin Impedance

Chapter 1

Introduction

1.1 Motivation

Parkinson's Disease (PD) is a progressive neurodegenerative condition marked by disturbances in the upper and lower motor control system, leading to symptoms such as resting tremors, stiffness, and slowness of movement, which impair daily functions and diminish quality of life. Existing monitoring techniques are expensive, reliant on hospitals, and inadequate for immediate or early identification. This initiative seeks to create a smart wristband for economical, ongoing monitoring and to aid therapy, filling current voids in PD management.

1.2 Aim of the project

The main technical goal of this project is to develop an intelligent wearable wristband ("TreCon") that can identify, predict, and help manage tremors through feedback stimulation. The particular responsibilities consist of:

- 1- Identification and Prediction: Employing surface Electromyography (EMG) and Inertial Measurement Units (IMU) to accurately identify and predict tremors associated with Parkinson's disease.
- 2- Incorporating Artificial Intelligence: Utilizing machine learning methods to detect initial tremor patterns and categorize sensor information.
- 3-Mobile Monitoring: Linking the device to a mobile app through Bluetooth for immediate monitoring and notifications.
- 4- Treatment Assistance: Utilizing electrical feedback stimulation to reduce tremor severity and enhance patient stability

1.3 Constraints

The creation and development of the TreCon wristband are influenced by various practical limitations:

- **Economic:** The device needs to be affordable to guarantee accessibility for regular users, in contrast to various proprietary commercial systems.
- **Health & Safety:** Being a wearable medical device, it needs to be safe for prolonged skin contact and during electrical stimulation.
- **Ethical considerations:** Ensuring patient data privacy and obtaining informed consent for the use of muscle and motion data are crucial.
- **Manufacturability:** The design needs to be compact and comfortable, appropriate for incorporation into one wearable solution.
- **Environmental:** The materials selected must be long-lasting for everyday use while considering their environmental effects

1.4 Standards:

The project adheres to various engineering and medical-device standards to guarantee signal precision, electrical safety, and dependable biomedical functionality. These consist of IEEE Std 1241-2010 concerning ADC testing and signal conversion, IEC 60601-1 related to medical electrical safety, IEC 60601-2-10 focused on nerve and muscle stimulator safety, SENIAM guidelines for surface EMG collection, and IEEE EMBS guidelines on biomedical signal processing.

These standards were primarily referenced in Sections 3.2 (System Design), 3.3 (System Analysis), and 3.4.8 (Safety Analysis).

1.5 Report Outline

Chapter 2 (Theoretical Background & Literature Review): Discusses the medical basis of Parkinson's disease and evaluates current wearable tremor monitoring technologies and methods for sensor fusion.

Chapter 3 (Design & Implementation): Outlines the hardware configuration (sensors, Arduino, ESP-12F), cloud connectivity (Firebase), Python-driven signal analysis (FFT and filtering), and the conceptual foundation for FES therapy.

Chapter 4 (Results & Discussion): Displays experimental results assessing the system's ability to differentiate between tremors and regular movement, along with data visualization for web and mobile platforms.

Chapter 5 (Conclusion & Future Work): Outlines the main accomplishments of the project and discusses upcoming research directions, such as implementing physical FES hardware, optimizing components, and updating software.

Chapter 2

Theoretical Background and Literature Review

2.1 Literature Review

Parkinson's Disease (PD) is a progressive neurodegenerative disorder characterized by the degeneration of dopaminergic neurons within the substantia nigra pars compacta, which disrupts the basal ganglia–thalamocortical motor circuit and produces classical motor symptoms including resting tremor, rigidity, and bradykinesia [1]. Tremor is particularly significant in PD, as it emerges from pathological oscillations between the basal ganglia, thalamus, and motor cortex, which propagate to peripheral muscles and can be captured through surface electromyography (sEMG) and inertial sensors [2]. The need for continuous, wearable, and clinically meaningful tremor detection systems has grown due to the fluctuating nature of symptoms and the limitations of hospital-based assessments that fail to capture real-world tremor variability.

Recent research highlights the role of surface EMG in detecting tremor-specific neuromuscular activity. Studies show that PD tremor typically presents in the frequency range of 4–7 Hz, making it distinguishable from voluntary movement, which commonly occurs above 8 Hz [3]. Hossen et al. proposed a wearable sEMG leg-band capable of capturing tremor episodes even before they become visually observable, demonstrating the sensitivity of EMG to subtle neuromuscular oscillations [4]. These findings support that EMG is a reliable physiological biomarker for tremor detection in both clinical and daily environments.

Complementing EMG, Inertial Measurement Units (IMUs) have been widely used for motion-based tremor analysis. Tzallas et al. demonstrated that IMU data, when processed with intelligent signal-analysis algorithms, can be used to quantify tremor severity and differentiate tremor from normal motion [5]. IMU-based solutions are portable, inexpensive, and practical for home monitoring. Furthermore, Delrobaei et al. used deep multiple-instance learning models on IMU data collected “in the wild,” successfully identifying tremor events with high accuracy despite

real-life noise and variability [6]. Their work validates that IMU sensors combined with AI are suitable for wearable tremor detection in uncontrolled environments.

Commercial wearable systems have also advanced the field. The Kinesia™ system integrates wearable accelerometers with clinical scoring algorithms to quantify tremor amplitude and frequency, providing FDA-cleared remote monitoring [7]. Similarly, the Verily Study Watch includes high-resolution IMUs for long-term neurological data collection [8]. While these devices are effective for tremor monitoring, they are often expensive, proprietary, and targeted mainly toward clinical research rather than everyday users seeking affordable solutions.

Non-invasive treatment-support technologies have emerged as promising complements to tremor detection. Peddle et al. introduced a mechanical tremor-suppression glove that cancels tremor motion by applying reactive forces [9]. Meanwhile, Pahwa et al. demonstrated that peripheral nerve stimulation using patterned electrical pulses can reduce tremor amplitude significantly in essential tremor (ET) patients [10]. These studies provide scientific evidence that wearable devices can not only detect tremor but also support non-pharmacological tremor reduction.

In addition to peer-reviewed journal studies, authoritative neuroscience and movement-disorder databases such as PubMed [11], ScienceDirect [12], and ACS Publications [13] provide extensive insights into PD pathology, tremor mechanisms, and recent advancements in wearable biomedical technologies. The peer-reviewed journal Tremor and Other Hyperkinetic Movements, which publishes studies on tremor classification, frequency characterization, and treatment outcomes, serves as a key knowledge base in this field [14].

Combining insights from these sources, the TreCon smart wristband project aims to develop a wearable device integrating surface EMG and IMU sensors for accurate tremor detection while offering a pathway for future non-invasive treatment support. The wearable will rely on signal-processing standards, AI algorithms, and biomedical safety guidelines to classify tremor events, differentiate them from voluntary motion, and transmit encrypted results to a mobile application. The reviewed literature strongly supports the feasibility of combining multimodal

sensors and intelligent algorithms in a compact wrist-worn device tailored for Parkinson's tremor detection.

Chapter 3

Design and Implementation

3.1 System Components

The system consists of various hardware and software elements that work together to enable real-time detection, analysis, and visualization of tremors [15],[22]. Every element is vital for maintaining the complete system's effectiveness and precision [15].

3.1.1 Hardware Components

The hardware layer is tasked with obtaining and sending physiological and motion signals for tremor recognition [22]. It combines sensing, processing, and communication components to guarantee precise data collection and dependable real-time operation [18],[19].

3.1.1.1 EMG Sensor

The Electromyography (EMG) sensor measures the electrical signals produced by muscle contractions. This signal delivers important insights into neuromuscular activity associated with tremor disorders [4],[25]. EMG signals are naturally weak and very prone to noise, making effective filtering and signal conditioning crucial for precise analysis [8],[26].

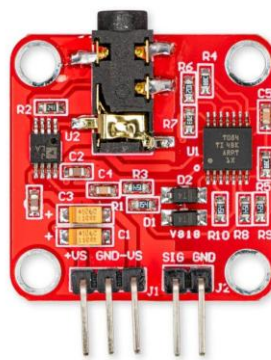


Figure (3-1) EMG Sensor [44]

3.1.1.2 Arduino Nano 33 BLE Sense Rev2

The primary data collection and embedded processing unit of the system is the Arduino Nano 33 BLE Sense Rev2 [18]. It incorporates an Inertial Measurement Unit (IMU) that delivers 9-axis motion detection, comprising a 3-axis gyroscope, 3-axis accelerometer, and 3-axis magnetometer.

This project employs solely the gyroscope [15],[23] component to gauge angular velocity and identify hand movements linked to tremors. Employing gyroscope data enables precise identification of oscillatory movement [6], a vital feature of tremor disorders.

The Arduino collects sensor data instantly, conducts preliminary preprocessing, and readies it for sending while maintaining signal quality.

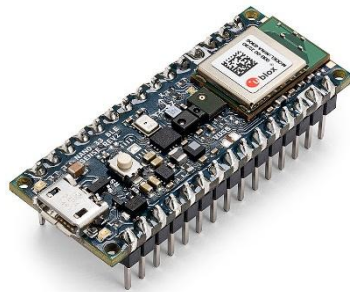


Figure (3-2) Microcontroller [45]

3.1.1.3 ESP-12F NodeMCU D1 Mini

The ESP-12F NodeMCU D1 Mini, built on ESP8266, functions as a specialized communication module for facilitating wireless data transfer [19]. It offers Wi-Fi access, enabling the system to transmit data to the cloud platform instantly [20]. This module was utilized since the Arduino Nano 33 BLE Sense Rev2 lacks Wi-Fi capability.

Nonetheless, this design presents a drawback, as employing a separate communication module raises system complexity and takes up extra physical space. Nonetheless, the distinction between the Arduino (data gathering and processing) and the ESP8266 (communication) enhances system modularity and reliability, enabling each component to execute its function effectively.



Figure (3-3) WiFi Chip [46]

3.1.2 Software Components

The software design includes cloud storage, data processing, and signal analysis modules. Firebase Realtime Database [20],[21] enables real-time data synchronization and remote access, while cloud-based IoT frameworks improve scalability and flexibility. The Arduino IDE is used to program the Arduino Nano 33 BLE Sense Rev2 for sensor interfacing and real-time data acquisition. A Python-based processing unit applies digital signal processing techniques [24], including filtering and feature extraction, to analyze the collected data and reduce noise [13].

3.1.3 Data Visualization Platform

The Blynk IoT platform is used for real-time visualization of processed tremor data [7], including amplitude, frequency, and severity levels. In addition to the mobile application, a web-based monitoring platform was developed to provide detailed analysis and historical data for healthcare professionals. These visualization systems improve usability and support continuous patient monitoring in biomedical applications [11].

3.1.4 Component Selection Criteria

The selection of system components is based on functionality, ease of integration, and cost efficiency. Additional considerations such as power consumption, availability, and scalability are also considered to ensure the system is practical and suitable for real-world applications [8].

3.2 System Design

The design of the system employs a modular and hierarchical framework, with each functional component working autonomously while aiding in the overall system functionality. This design method enhances system adaptability, upkeep, and potential growth. Furthermore, the design is created in line with applicable IEEE and ISO standards [7],[9] to guarantee reliability, consistency, and adherence to engineering best practices.

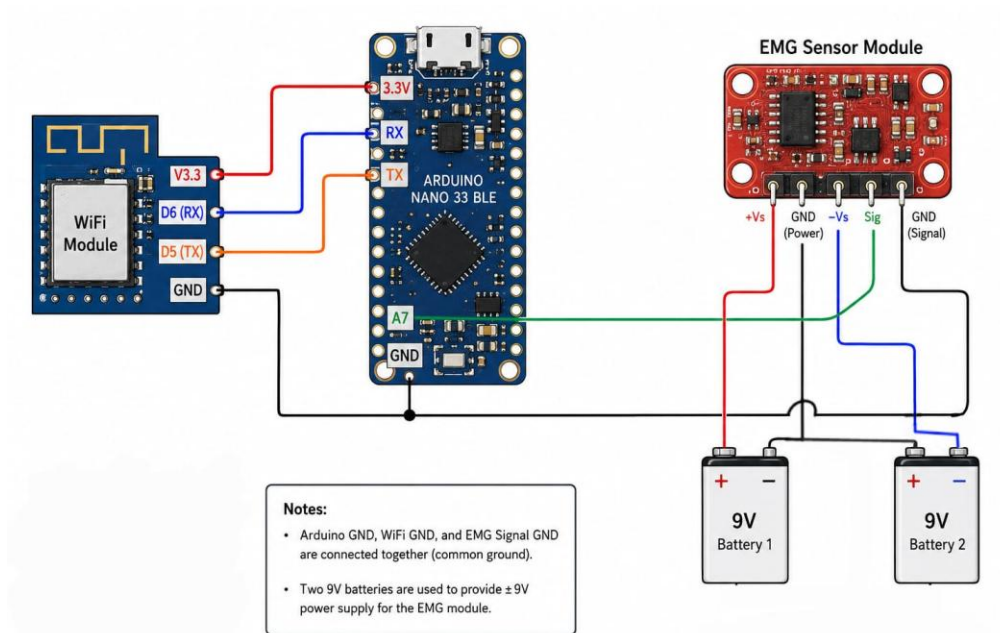


Figure (3-4) Circuit Connection Diagram

3.2.1 Circuit Design and Electrical Analysis

The circuit configuration integrates the EMG sensor, Arduino Nano 33 BLE Sense Rev2, and ESP-12F NodeMCU D1 Mini [9],[19] to facilitate immediate collection, processing, and wireless transmission of tremor data. The entire configuration follows a modular structure, where sensing, processing, and communication are interconnected both electrically and functionally.

3.2.1.1 EMG Sensor Electrical Operation

The EMG sensor utilizes a differential input setup to capture bio-potentials[25],[28]. While the system logic is powered via a laptop USB connection, the sensor is powered by an independent dual-rail supply from two 9V batteries. This provides the necessary $+V_s = +9V$, GND, and $-V_s = -9V$ rails specified in the Muscle Sensor v3 User's Manual.

Applying Kirchhoff's Voltage Law (KVL) to the sensor supply loop confirms the total potential difference: $\sum V = (+9V) - (-9V) = 18V$

The manual specifies that differential input signals usually range between 2 mV and 5 mV [25]. The sensor amplifies, rectifies, and smooths this signal into an envelope output (V_{out}). Per the datasheet, the output range is: $V_{out} \in [0, +V_s]$ So, $0V \leq V_{out} \leq 9V$.

The sensor's gain is defined by the resistance of the onboard potentiometer: $Gain = 207 * \frac{R}{1k\Omega}$

Using manual's typical value of $R = 50 k\Omega$: $Gain = 207 * \frac{50k\Omega}{1k\Omega}$

This high gain enables the amplification of weak muscle signals for processing [26].

3.2.1.2 Arduino Nano 33 BLE Sense Rev2 Interface

The Arduino Nano 33 BLE Sense Rev2 operates at a 3.3V logic level [18], regulated from the laptop's USB 5V supply (ABX00069-datasheet). It features a 12-bit ADC [27] where the resolution is determined by the reference voltage $V_{ref} = 3.3V$ and the bit depth $2^{12} = 4096$. $Resolution = \frac{3.3v}{4096} = 0.805 mV$

The EMG signal is connected to analog pin A7. Because the system works properly without external resistors, the signal is sampled directly. According to Ohm's Law ($V = I * R$), and given the high input impedance of the nRF52480 ADC (*typically* $1M\Omega$), the current drawn from the sensor is: $I = \frac{3.3 v}{1M\Omega} \approx 3.3 \mu A$

This minimal current ensures a negligible loading effect [27], preserving the integrity of the 10,350x amplified signal.

3.2.1.3 IMU (Gyroscope) Signal Acquisition

The integrated IMU offers 9-axis detection, featuring a 3-axis gyroscope, accelerometer, and magnetometer. In this setup, solely the gyroscope is utilized.

The gyroscope functions within adjustable scopes: $\pm 125, \pm 250, \pm 500, \pm 1000, \pm 2000$ °/s

The angular velocity is represented as: $\omega = \Delta\theta / \Delta t$

This data is used to detect tremor frequency and intensity [6].

3.2.1.4 ESP8266 Communication Circuit

The ESP-12F NodeMCU D1 Mini operates at 3.3V with a clock frequency of 80 – 160 MHz and supports IEEE 802.11 b/g/n Wi-Fi communication.

Communication between Arduino and ESP8266 is achieved via UART (TX/RX). Since both devices operate at 3.3V logic, no level shifting is required [18],[19].

Power consumption can be estimated as: $P = V \times I$

3.2.1.5 Power Distribution and Circuit Analysis

The system uses two 9V batteries to supply the EMG sensor with a dual voltage configuration (+9V, GND, -9V). Currently, an external regulated power supply is used to provide a stable 3.3V input [18],[19] for the Arduino Nano 33 BLE Sense Rev2 and the ESP8266 module to ensure reliable operation during testing and development. However, the overall design goal is to integrate all system components into a unified battery-powered configuration for full portability [22].

Using Voltage Divider Rule (VDR): $V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$

This ensures stable voltage levels when needed.

Using Current Divider Rule (CDR): $I_k = I_{tot} \times \frac{R_{tot}}{R_k}$

This helps in analyzing current distribution across parallel paths.

Applying Kirchhoff's Current Law (KCL): $\Sigma I_{in} = \Sigma I_{out}$

This ensures stable current flow across the circuit nodes.

3.2.1.6 System Capability and Performance Evaluation

Arduino's 64 MHz processor and 12-bit ADC provide sufficient computational power and resolution for biomedical signal processing.

The EMG sensor provides high gain and signal conditioning [25], enabling detection of weak muscle signals. The ESP8266 ensures reliable wireless communication [20] and IoT integration with cloud platforms. Overall, the system demonstrates high capability in real-time tremor monitoring [22], balancing accuracy, processing speed, and communication efficiency

3.2.2 Physical Design

The physical implementation of the system focuses on compactness, portability, and user comfort [15]. All electronic components are assembled on a stripboard to ensure stable connections and efficient space utilization. Additionally, a custom 3D-printed enclosure is used to securely house the components while providing protection and maintaining a lightweight, wearable design

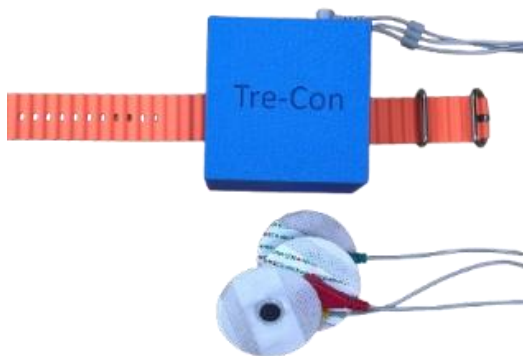


Figure (3-5) 3D Printed Design

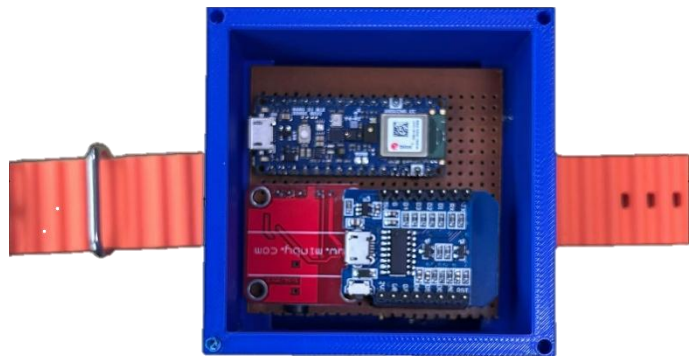


Figure (3-6) Stripboard Design

3.3 System Analysis

3.3.1 Sampling Theory

Biomedical signals were digitally captured [26],[27] to retain physiological data without aliasing or distortion. As per the Nyquist sampling theorem, the sampling rate needs to meet: $f_s \geq 2f_{max}$

Because Parkinsonian tremors happen within the frequency range: $3Hz \leq f_{tremor} \leq 8Hz$

the least necessary sampling frequency turns into: $f_s \geq 2(8) = 16Hz$

To enhance FFT precision and frequency clarity, the chosen sampling frequency for this project was: $f_s = 20Hz$

The chosen sampling rate meets the Nyquist criterion [24],[26] and offers adequate temporal resolution for dependable tremor identification and spectral analysis.

3.3.1.1 Analog-to-Digital Conversion

The analog EMG signal is transformed into digital data [27] through the internal ADC of the Arduino Nano 33 BLE Rev2 for additional processing and analysis.

The ADC procedure quantizes the input signal [27] based on the resolution of the converter. The digital depiction is provided by: $\text{Digital Value} = \frac{V_{in}}{V_{ref}} (2^n - 1)$

3.3.2 EMG Muscle Sensor Analysis

Arduino-gathered surface EMG signals capture muscle activity [25] alongside noise; digital signal processing is employed to remove disturbances for clearer, more accurate readings.

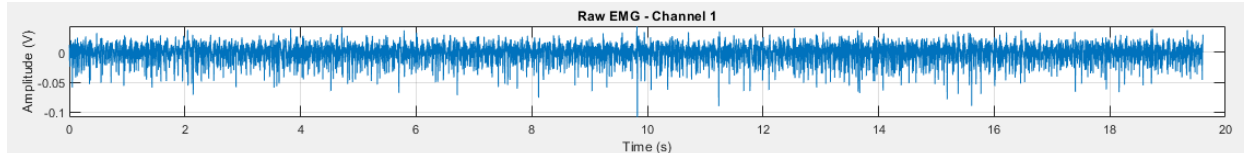


Figure (3-7) Generated EMG Data Before Analysis

3.3.2.1 Noise Sources

Biomedical signals are susceptible to noise and artifacts that can affect signal quality and tremor detection, including various interference sources during acquisition.

1. Power line interference.
2. Motion artifacts.
3. Electrode instability.
4. Sensor drift.
5. High-frequency environmental noise.

3.3.2.2 EMG Band-Pass Filtering

A digital Butterworth band-pass filter was utilized [24] to extract the specific tremor frequencies while minimizing undesired elements. The selected frequency range was: $4\text{Hz} \geq f_s \geq 12\text{Hz}$

The normalized cutoff frequencies are calculated as: $W_n = \frac{f_c}{f_s/2}$

A fourth-order Butterworth filter was chosen [26] due to its consistent behavior and smooth frequency response. The filter transfer function is represented by: $H(s) = \frac{1}{\sqrt{1 + (\frac{s}{w_c})^{2n}}}$

3.3.2.3 EMG Rectification

MG signals are rectified to change all values to positive for simpler analysis.

The corrected signal is derived by utilizing: $EMG_{\text{rect}}(n) = |EMG(n)|$

This method transforms all negative values into positive amplitudes while maintaining the characteristics of signal energy.

3.3.2.4 RMS Envelope Extraction

RMS measured muscle activity and signal power, creating a smooth envelope that indicates activation intensity. The RMS value is determined by using $RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$

A 200 ms interval was utilized for real-time RMS computation to maintain a balance between quick response and smooth signal output.

3.3.2.5 Fast Fourier Transform (FFT)

FFT was applied to EMG signals following rectification and RMS processing to assess muscle activity and identify tremors.

The Discrete Time Fourier Transform is represented as: $X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$

The dominant tremor frequency is obtained from [3]: $f_{\text{dominant}} = f(K_{\text{max}})$, $k_{\text{max}} = \text{argmax}|X(k)|$

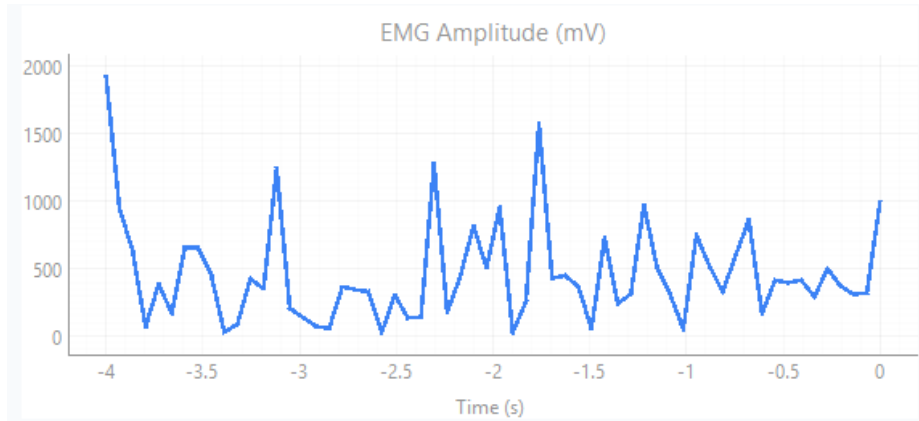


Figure (3-8) Final Output EMG Signal

3.3.3 IMU Digital analysis

The IMU gyroscope detects angular velocity [15],[18] across the x, y, and z axes, and the information is processed through digital techniques such as FFT

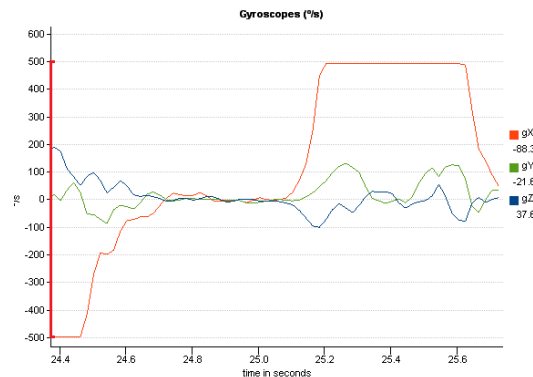


Figure (3-9) Individual 3 Axis IMU [47]

3.3.3.1 Introduction to IMU-Based Tremor Monitoring

Parkinsonian tremors happen at a frequency of 3–8 Hz, and IMU wearables facilitate real-time, non-invasive detection of these tremors.

To ease the digital signal analysis, the three axis values have merged into a single vector based on:

$$M = \sqrt{X^2 + Y^2 + Z^2}$$

3.3.3.2 IMU Signal Characteristics

The gyroscope signal represents overall hand movement, so analyzing only one axis may not fully capture tremor activity.

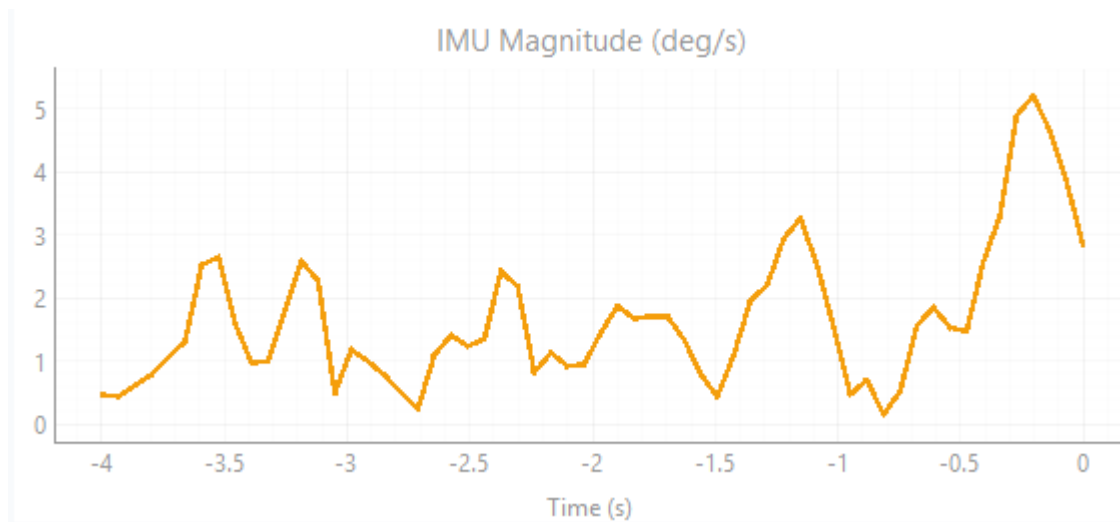


Figure (3-10) Gyroscope graph result

The vector mentioned above has been created through a combination process that produces a single waveform indicating the overall intensity of angular movement, irrespective of the direction.

3.3.3.3 Noise Sources in IMU Signals

IMU signals have noise that may lower detection precision, and the low-frequency characteristics of Parkinson's tremors can influence FFT analysis. $IMU_{raw}(t) = IMU_{tremor}(t) + noise(t)$.

1. Motion Artifacts: IMU noise and low-frequency tremors reduce FFT-based detection accuracy.
2. Sensor Drift: Sensor drift causes slow gyroscope errors that distort long-term motion signals.
3. Quantization Noise: ADC sampling converts IMU signals into discrete values, introducing quantization noise.

Furthermore, sensor Orientation Variation noise arises when the IMU device is not securely held in the hand.

3.3.3.4 IMU Digital Filtering

Digital filtering replaced analog filters to enhance IMU signal quality by eliminating noise, undesired frequencies, and environmental interference, while isolating tremor-related oscillations and improving FFT accuracy with a better signal-to-noise ratio. A band-pass filter ranging from 1 to 12 Hz was developed based on the characteristics of Parkinsonian tremor in biomedical studies, where the lower cutoff (1 Hz) minimizes slow drifts and external noise, while the upper cutoff (12 Hz) eliminates higher-frequency motion artifacts. The system adhered to the Nyquist sampling criterion ($f_s \geq 2f_{max}$) to avoid aliasing, and typical biomedical preprocessing methods were utilized to guarantee clean and dependable motion signal conditioning prior to frequency-domain analysis.

3.3.3.5 Butterworth Band-Pass Filter

The IMU gyroscope signal includes noise and extraneous frequencies due to motion, electrical interference, and sensor instability, potentially impacting tremor analysis. A Butterworth band-pass filter with a range of 1–12 Hz was utilized to isolate frequencies related to Parkinsonian tremors, eliminating other elements and emphasizing rotational motion instead of EMG filtering. Where the processed IMU signal is depicted as:

$$IMU_{filtered}(t) = \text{BandPass}(IMU_{raw}(t))$$

A fourth-order Butterworth filter was employed, akin to EMG analysis, due to its stable response and smoothness, reducing distortion while maintaining oscillatory motion signals.

3.3.3.6 Fast Fourier Transform (FFT) Analysis

FFT is utilized two times: initially for filtering and then to determine tremor frequency from rotational movement. The resulting frequency serves as a criterion for identifying and categorizing the presence of tremors according to a set threshold.

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

Employing the FFT, the frequency associated with the highest spectral amplitude is obtained based on: $f_{IMU,max} = f(K_{max})$ and K_{max} is defined as: $K_{max} = \text{argmax}(|X(k)|)$.

3.3.3.7 IMU Tremor Detection Contribution

Below are the frequency ranges that is defined for our design system:

- For normal Parkinsonian Tremor Detecting: $8\text{Hz} \geq f_{IMU,FFT} \geq 4\text{Hz}$ [2],[23]
- For high/harmful Parkinsonian Tremor Detecting: $25\text{Hz} \geq f_{IMU,FFT} \geq 8\text{Hz}$

A threshold of 4 Hz is set to differentiate tremors

from regular movement, since voluntary actions generally happen beneath this frequency. Frequencies greater than 4 Hz suggest potential Parkinsonian tremors, and the same FFT technique used in EMG analysis is utilized

3.3.4 Tremor Detection Algorithm

Thresholds that adapt according to RMS statistics were employed, and a tremor alert activates only when both IMU and EMG signals identify tremors using an AND logic condition Below is the IMU threshold setpoint: $\text{Magnitude}_{FFT} \geq 0.08$

If the magnitude of the IMU surpasses a predetermined experimentally set threshold, it indicates a tremor; if not, the system categorizes it as no tremor

The tremor condition is defined as:

$$\text{Tremor} = (\text{Magnitude}_{FFT} > f_{\text{threshold}}) \wedge (f_{IMU} > \text{Threshold}) \wedge (f_{EMG,FFT} > f_{\text{threshold}})$$

3.3.5 Firebase and Cloud Monitoring

Wireless communication was deployed to enable immediate remote observation of patient conditions. Processed data, including tremor level, EMG activity, and dominant tremor frequency, were transmitted wirelessly to cloud visualization platforms. The Python Website along with the Blynk IoT platform was utilized to display the collected signals and monitoring parameters via graphical dashboards. Real-time graphs enable ongoing monitoring of patient states and could facilitate future telemedicine solutions.

The parameters being conveyed consist of:

- IMU measurements.
- Tremor status if mid\high\low
- EMG measurements.
- Tremor frequency.

3.3.6 Coding

The Arduino IDE facilitated the programming of the Arduino Nano 33 BLE Rev2 and the ESP8266 Wi-Fi module for gathering sensor data and wirelessly sending it to the Firebase cloud database. Digital signal processing involved the use of Python for tasks such as filtering, FFT analysis, tremor detection, and decision-making utilizing IMU and EMG data. Moreover, Python was utilized to create a graphical user interface (GUI) for real-time monitoring

3.4 Tremor Treatment

Pathological tremors impact life quality; medications can decline in effectiveness, and Deep Brain Stimulation is invasive, costly, and risky [31],[32],[33].

3.4.1 Focused Ultrasound (MRgFUS)

Targeted ultrasound therapy uses high-intensity ultrasound waves guided through the skull to precisely target brain regions, while MR thermometry monitors temperature in real time for controlled treatment. This technique has shown significant long-term tremor reduction in Parkinson's patients with minimal serious procedure-related complications [34],[35].

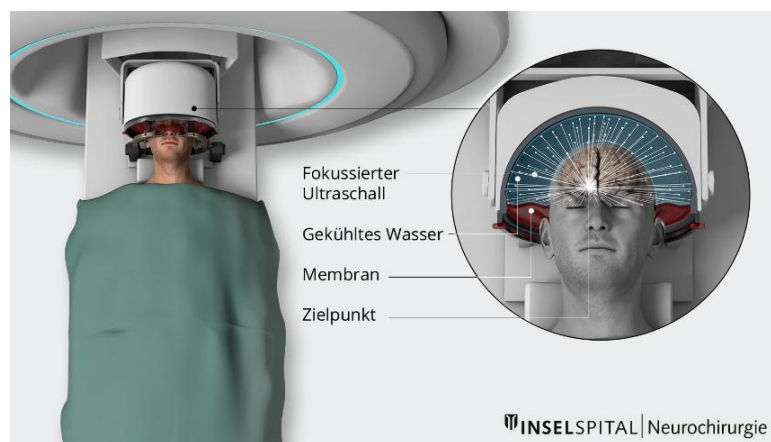


Figure (3-11) Focused Ultrasound Device [48]

3.4.1.1 Critical Limitations

HiFUS (MRgFUS) can reduce tremor symptoms, but patient outcomes vary and only about half experience long-term improvement. The procedure is permanent, limited by skull density, and as an open-loop system, it cannot adapt to tremor changes or disease progression over time.

3.4.2 Functional Electrical Stimulation (FES)

Unlike lesion-based therapies, Functional Electrical Stimulation (FES) [33] is a non-invasive and wearable technique that uses electrical stimulation to produce controlled muscle contractions and counteract involuntary tremor movements in real time [9],[10],[36].



Figure (3-1) FES Stimulation Pads [49]

3.4.2.1 Destructive Interference of Biomechanical Waves

The pathological tremor $T_{\text{Tremor}}(t)$ is modeled as a narrowband, quasi-sinusoidal oscillation with time-varying parameters. The FES system generates an anti-tremor force $T_{\text{FES}}(t)$ with matched amplitude $A(t)$ and frequency $f(t)$ but a phase shift of exactly π radians (180°). The resultant net torque on the target joint is: $T_{\text{Net}}(t) = T_{\text{Tremor}}(t) + T_{\text{FES}}(t)$

When the control objective is achieved by $T_{\text{FES}}(t) = -T_{\text{Tremor}}(t)$ – then $T_{\text{Net}}(t) \approx 0$, yielding effective tremor cancellation.

3.4.2.2 Anti-Phase Stimulation

The principle of using FES for tremor suppression is not to simply activate muscle, but to use a specific control strategy known as out-of-phase or anti-phase stimulation [37],[38].

3.4.2.2.1 Tremor Signal Model

The filtered tremor signal from the IMU, after sensor fusion, is modeled as a sinusoid with time-varying parameters: $Tremor(t) = A_{tremor}(t) * \sin(\omega_{tremor}(t) * t + \phi_{tremor}(t))$ [3],[24]

3.4.2.2.2 Anti-Phase Control Law

The stimulation command force is defined as:

$$F_{anti,cmd}(t) = G * A_{tremor}(t) * \sin(\omega_{tremor}(t) * t + \phi_{tremor}(t) + \pi + \phi_{compensate})$$
 [37],[38]

3.4.2.2.3 Electromechanical Delay (EMD) Compensation

EMD is the fixed latency between electrical stimulation pulse application and the onset of measurable mechanical force [39],[42]. It is the sum of three components:

$$EMD = t_{chemical} + t_{excitation-contraction} + t_{series-elastic}$$

Table3-1 Muscle Group

Muscle Group	Fiber Type Predominance	EMD Range (ms)
Wrist flexors/extensors (FCR, ECR)	Mixed (Type I/II)	20-30
Forearm pronators / supinators	Mixed	25-35
Finger flexors (FDS, FDP)	Fast twitch (Type II)	15-25

Problem Quantification: For $EMD = 30$ ms and $f_{tremor} = 6$ Hz (period $t = 167$ ms), the uncompensated phase error is: $\Delta\phi_{uncomp} = (EMD / T_{tremor}) * 360^\circ = (30 \text{ ms} / 167 \text{ ms}) * 360^\circ \approx 65^\circ$. A 65° phase error reduces the efficiency of tremor cancellation from 100% (at 180°) to approximately $\cos(65^\circ) \approx 42\%$ of maximum possible suppression.

Solution: Phase Pre-Advance Compensation

The algorithm estimates EMD and determines when to start/stop FES for anti-phase cancellation

$$\phi_{cmd} = \phi_{tremor} + \pi + (\omega_{tremor} * EMD_{estimated})$$
 [37],[42]

3.4.3 Control System Architecture

The treatment module operates as a slave in a deterministic, real-time master-slave control architecture [33]. The master is the Analysis Module (sensor fusion) [38]. The slave is the Treatment Module (FES actuator). The total loop latency from tremor detection to applied counterforce must satisfy: $T_{loop} = T_{sense} + \tau_{compute} + T_{EMD} + T_{stimulation} \leq 10 \text{ ms}$ [47],[42]

This requirement stems from the need to track a 4-12 Hz pathological tremor signal (period $t = 83\text{-}250 \text{ ms}$) [3],[33]. A loop delay exceeding 10 ms introduces a phase error $\Delta\phi = \omega_{tremor} * T_{loop} \geq 18^\circ$ at 5 Hz, which degrades cancellation efficiency.

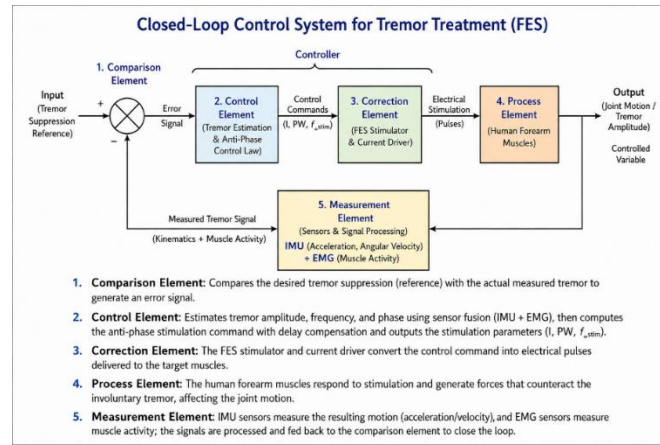


Figure (3-2) Closed loop System

3.4.3.1 Signal Chain Stages

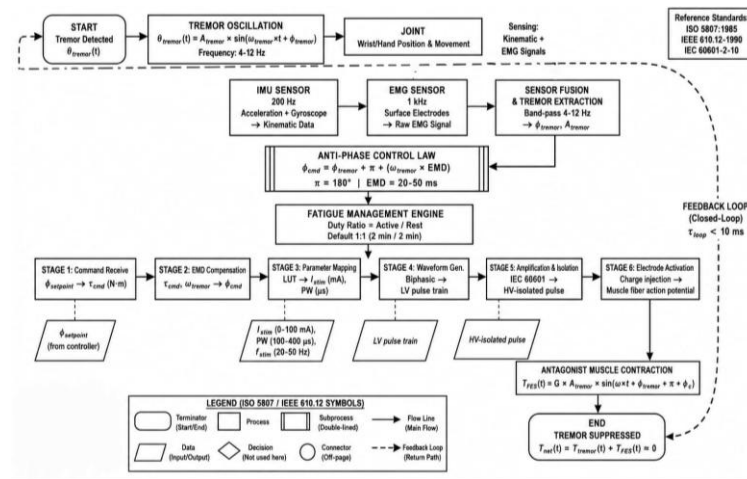


Figure (3-3) Signal Chain Stages

3.4.4 Stimulation Parameters

The FES stimulator has three adjustable parameters controlled by the algorithm.

Table 0-2 Important Parameters

Symbol	Typical Range	Resolution	Physiological Effect	Engineering Constraint
I [39]	0-100 mA	1mA	More motor units increase force.	Higher → faster fatigue
PW [42]	100 - 400 μs	1 μs	Higher charge recruits deeper fibers.	Wider → higher charge density
f_{stim} [42]	10 - 50 Hz	1Hz	Controls force smoothness.	Low → less fatigue, rough F; high → smooth F, more fatigue.

3.4.5 Hardware Architecture for Selective Activation

3.4.5.1 High-Density Electrode Matrix and Mapping System

High-density electrode arrays enhance selective finger control in FES systems [15],[33] by boosting spatial resolution across forearm motor points . Automated motor point mapping systems improve electrode positioning [41] by sequentially stimulating electrodes, capturing movement responses, and creating precise motor maps without manual trial-and-error methods.

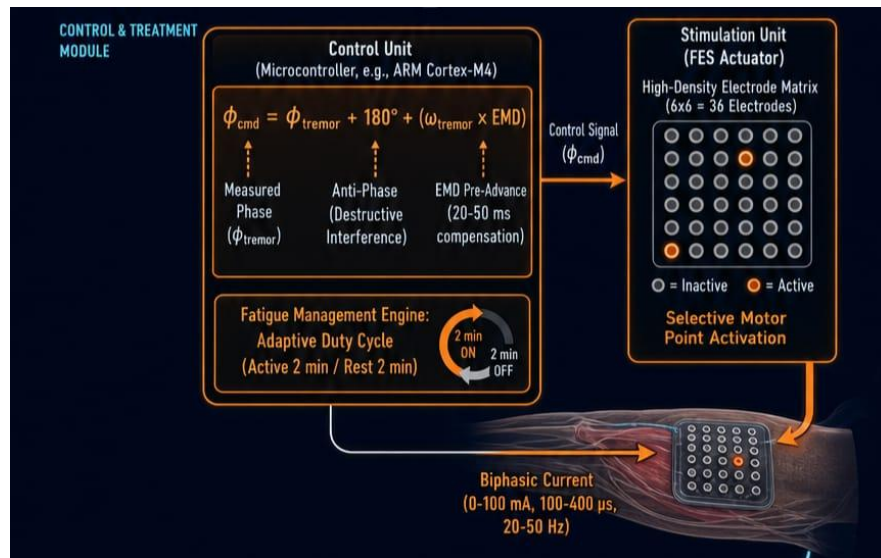


Figure (3-4) Control and Treatment Module

3.4.6 Why FES ?

FES was chosen because it is a low-cost, non-invasive, wearable, and closed-loop system [31],[36] that provides effective tremor suppression while remaining fully reversible and capable of supporting neuroplasticity. Although DBS can achieve higher suppression rates and MRgFUS is also non-invasive, both methods have major limitations such as invasiveness, irreversible effects, high cost, or lack of adaptability. In contrast, FES offers a safer, adaptive, and more practical solution for long-term tremor management [38].

3.4.7 The Fatigue Problem

Muscle fatigue is the primary limitation of FES for tremors [42]. Continuous, unbroken stimulation for hours is unsafe and ineffective [43].

3.4.7.1 Wear Pattern Recommendations

Continuous electrical stimulation can cause tissue heating and muscle fatigue [39], making it unsafe for long-term use. Intermittent duty-cycled stimulation with rest periods is safer [41] and more effective, while structured session-based protocols are clinically preferred and improve symptoms such as bradykinesia and gait.

3.4.7.2 Fatigue Onset Data

Table 0-3 Force Drop for Several Frequency [42]

Stimulation Frequency	Muscle	Time to 50% Force Drop
35Hz	Vastus Lateralis (fast twitch)	~14 minutes
50Hz	Vastus Lateralis (fast twitch)	~12 minutes
10Hz	Vastus Lateralis	>30 minutes (minimal fatigue)
35Hz	Quadriceps (paralyzed)	Model-predictable decay

3.4.7.3 Adaptive Duty Cycle Formula

Duty_Cycle_Ratio = Active_Time / Rest_Time [39]. Start with 1:1 (e.g., 2 min active / 2 min rest).

Adjust based on:

- Real-time EMG median frequency monitoring (decreases with fatigue) [25]
- Force output decay tracking [42]
- Patient-reported comfort [33]

3.4.8 Safety Analysis

3.4.8.1 Absolute & Relative Contraindications

FES has absolute and relative contraindications [40] that must be evaluated for safe use. Absolute contraindications include conditions such as pacemakers, uncontrolled epilepsy, and pregnancy [36], while relative contraindications like skin problems, cardiac conditions, or recent implants require careful assessment and supervision [40].

3.4.8.2 Hardware Safety Interlocks

Table 0-4 Protection Considerations [29],[40]

Protection Feature	Method	Trip Condition	Action
Overcurrent	Hardware comparator	$I_{stim} > 105 \text{ mA}$ (any pulse)	Instant crowbar shutdown & latch.
Pulse width limit	Hardware one-shot	$PW > 500 \mu\text{s}$ (any pulse)	Pulse truncated, device fault flag
Charge density limit	Software + hardware	Exceeds $30 \mu\text{C}/\text{cm}^2/\text{phase}$	Reduce amplitude/PW, alert user
Lead-off detection	Impedance measurement	$Z_{skin} > 20 \text{ k}\Omega$ for $> 100 \text{ ms}$	Shut down output, user alert
Watchdog timer	Independent timer	No command received for $> 50 \text{ ms}$	Reset MCU, shut down output
Thermal monitor	NTC thermistor	Electrode temp $> 41^\circ\text{C}$	Shut down, cool-down period

Chapter 4

Results and Discussion

4.1 What the Device Is Able to Do

The TreCon prototype accomplished real-time tremor detection by integrating EMG and IMU sensing with signal processing, cloud transmission, and visualization. Despite functional electrical stimulation not being integrated into the hardware because of the situation, all other components of the system were validated successfully. The hardware has been completely assembled.

4.2 Results Match Parkinsonian Values from Literature

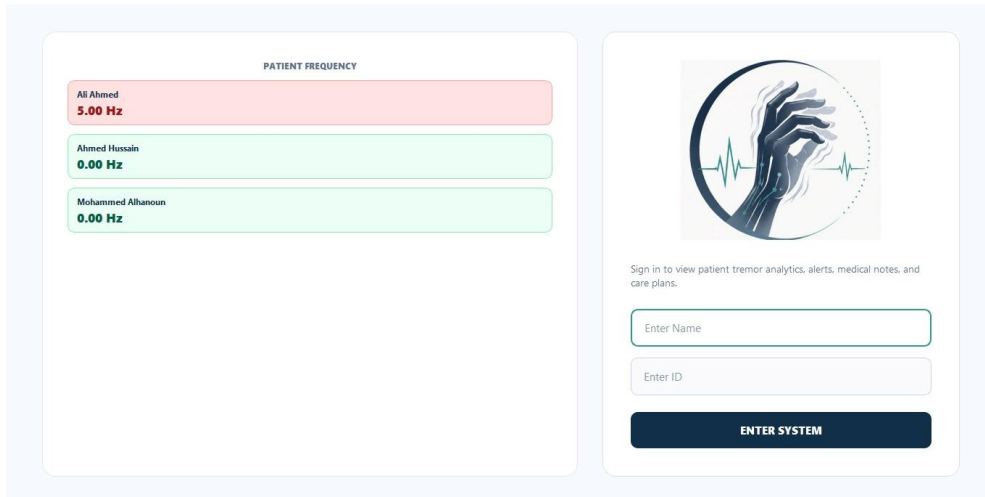
The apparatus was evaluated for the detection of Parkinsonian tremors (4–7 Hz). Tremor signals reliably stayed within this range, whereas voluntary movements exhibited higher frequencies, validating precise differentiation through FFT analysis of EMG and IMU data consistent with current literature.

4.3 Detection and Decision Accuracy

The device applies an AND logic condition where EMG and IMU FFT magnitudes have to be greater than 0.08 and the frequency needs to be between 4 and 12 Hz for tremor detection. This minimizes false positives and has been confirmed to effectively differentiate tremors from typical movement.

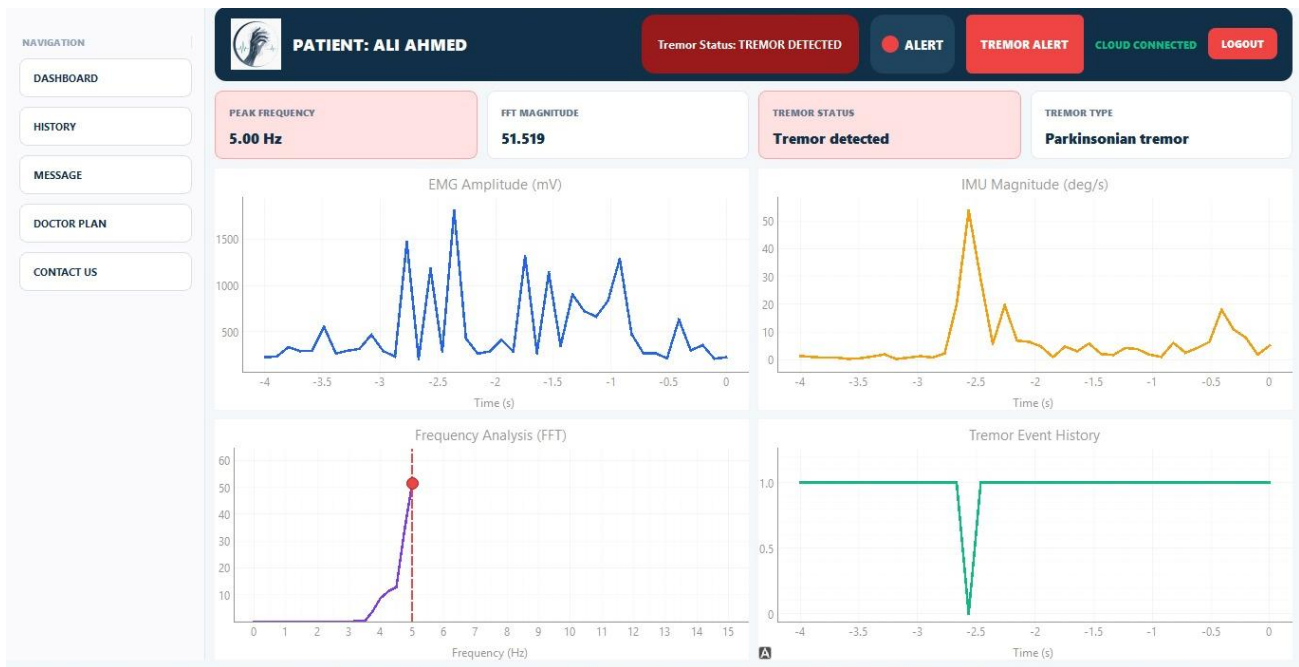
4.4 Clinical Web Dashboard (Doctor Perspective)

A web dashboard was created using Python to showcase the remote monitoring feature. It offers secure user access, and real-time visuals are presented for physicians addressing the challenge of capturing real-world tremor variability. In addition, the patient frequencies appears so the doctor notice the variation from the main screen.



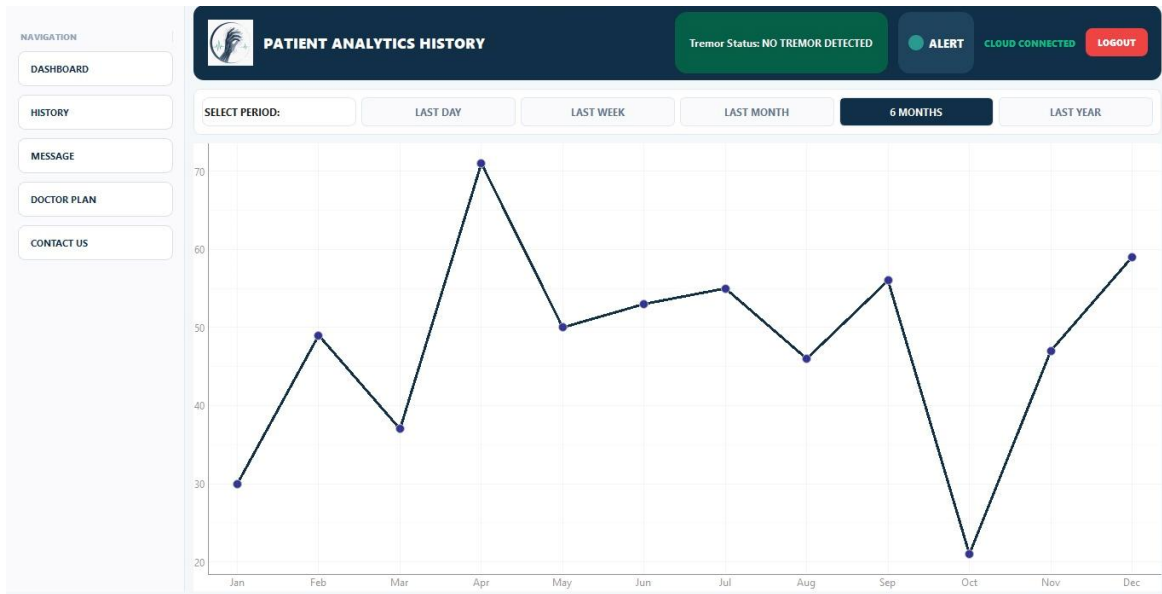
Figure(4-1) Website Main Page

View a dashboard displaying essential charts and patient metrics for observation. It features an automatic notification that triggers when a tremor is sensed for quick action.



Figure(4-2) Website Dashboard Section

Alongside, there is a history section where doctors can see how many tremors have been recorded over the selected time.



Figure(4-3) Website History Section

Messaging capability enabling doctors to reach out to patients directly, with correspondence sent via email for smooth communicationDash

The screenshot shows the 'MEDICAL MESSAGES' section. The navigation menu on the left is the same as in Figure 4-3. The header includes the same tremor status and alert indicators. The main content area is divided into two columns. The left column, titled 'PAST MEDICAL NOTES', contains a list of notes: '2026-05-01: Patient showing resting tremor.', '2026-05-04: Increase Sinemet dosage.', and '2026-05-08: Improved mobility.'. The right column, titled 'WRITE NEW RECOMMENDATION', has a text input field with the placeholder 'Type clinical observation...'. At the bottom of this column is a blue button labeled 'SEND TO RECORD'.

Figure(4-4) Website Messaging Feature

Section for treatment plans where physicians can record and oversee each patient's treatment strategy for systematic medical documentation and follow-up.

The screenshot shows the 'DOCTOR PLAN' section of a web application. On the left is a navigation sidebar with buttons for DASHBOARD, HISTORY, MESSAGE, DOCTOR PLAN (which is highlighted), and CONTACT US. The main content area has a dark blue header with a logo, the title 'DOCTOR PLAN', and status indicators: 'Tremor Status: NO TREMOR DETECTED', an 'ALERT' button, 'CLOUD CONNECTED', and a 'LOGOUT' button. Below the header, there are two main panels. The left panel, titled 'PLAN & MEDICATION HISTORY', contains a text input field with the value 'Plan: Levodopa 100/25 mg daily'. The right panel, titled 'NEW CLINICAL PLAN', contains a text input field with the placeholder 'Type new clinical plan...'. At the bottom right of the main content area is a green button labeled 'UPDATE PATIENT PLAN'.

Figure (4-5) Website Doctor Plan Section

The Contact Us section offers the doctor's information, hospital details, and emergency phone numbers for convenient communication.

The screenshot shows the 'CONTACTS' section of the web application. It features the same navigation sidebar as the previous section. The main content area has a dark blue header with the title 'CONTACTS' and the same status indicators. Below the header, there are several text input fields for contact information. The first field is labeled 'DOCTOR FULL NAME' and contains the text 'Dr. XXXXXXX, Senior Neurologist'. The second field is labeled 'OFFICE PHONE' and contains the text '+973 17XX XXXX'. The third field is labeled 'HOSPITAL NAME' and contains the text 'XXXXX Clinical Research Hospital'. The fourth field is labeled 'EMERGENCY UNIT' and contains the text 'Ext. 999'.

Figure (4-6) Website Contact Us Section

4.5 Blynk IoT Application (Patients and Families)

A Blynk IoT mobile application was developed to provide simple monitoring for patients and their families. It focuses on real-time accessibility rather than clinical complexity. It provides live sensor updates and a clear tremor indicator for easy interpretation by non-experts.

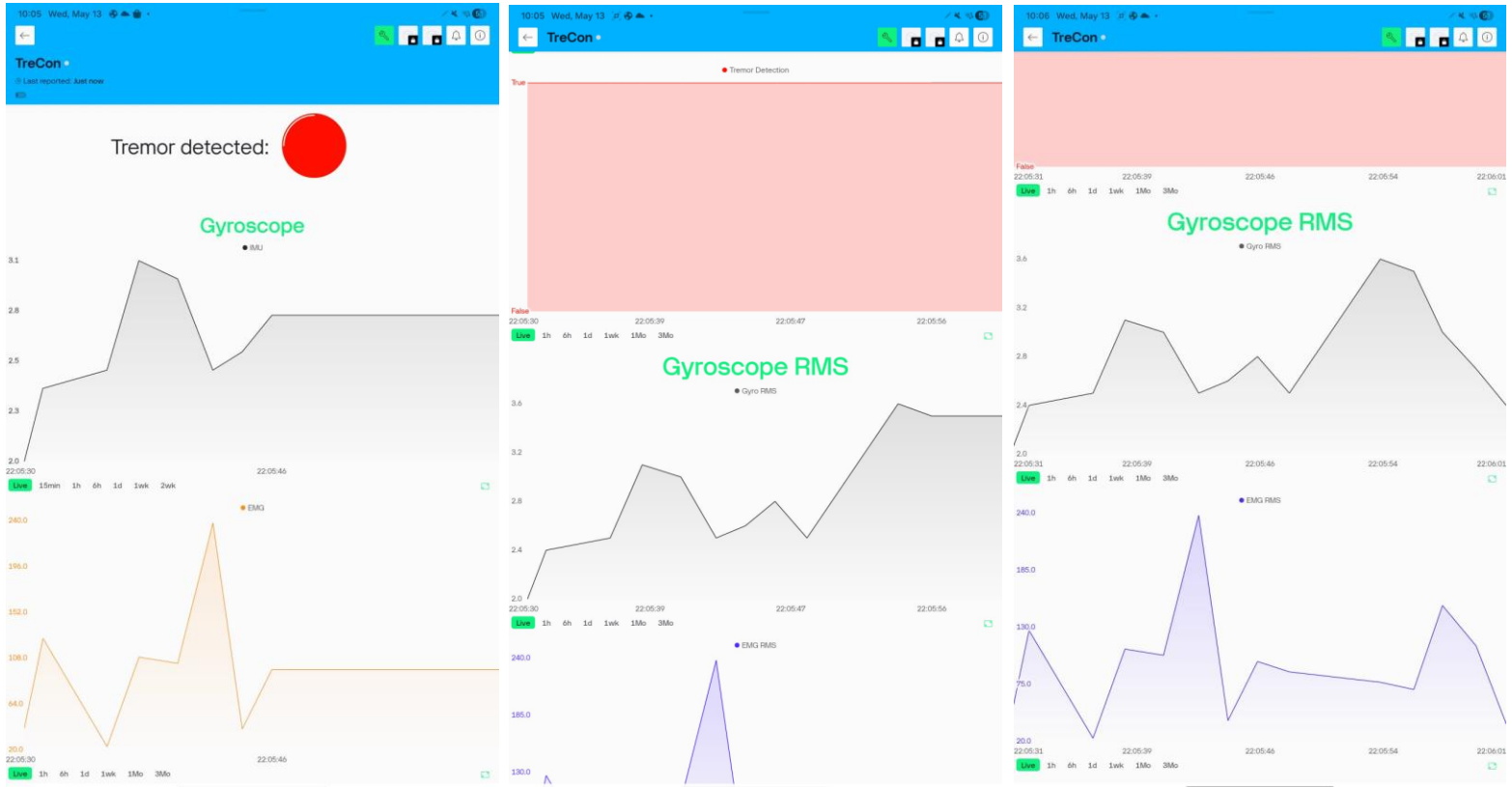


Figure (4-7) Application Dashboard

4.6 Evidence That the Problem Was Solved

The project solves the need for an affordable, portable, real-time tremor monitoring system. With a low cost of about 55 BHD, it offers a cheaper alternative to commercial systems. Although treatment (FES) was not implemented as hardware, all monitoring and detection goals were successfully achieved plus treatment research proving it works.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

The Tre-Con initiative tackles the difficulties of observing Parkinson's Disease (PD) tremors by creating an intelligent wearable wristband for ongoing, real-time monitoring beyond clinical environments. By integrating surface Electromyography (sEMG) with an Inertial Measurement Unit (IMU), the system generates a detailed profile of muscle function and movement dynamics.

Employing sophisticated digital signal processing—namely Butterworth filtering and FFT analysis using python the device effectively differentiates Parkinsonian tremors from voluntary movements on the device. Furthermore, the study creates a theoretical model for a closed-loop suppression system incorporating non-invasive Functional Electrical Stimulation (FES).

5.2 Future Work

Given that the present stage of the project emphasizes research and the creation of the signal processing framework, the main goal for future efforts is the complete hardware realization of the FES treatment module. Integrating the physical stimulation circuitry will shift the device from just being a monitoring tool to a comprehensive assistive treatment solution. To improve the device's usability, future versions should choose specialized, high-precision hardware parts that take up less space, facilitating a more compact and user-friendly 3D-printed casing. Moreover, integrating contemporary AI models for enhanced predictive analysis and broadening the influence of web and mobile applications like including patient history trends and automated clinician reports will enhance the system's effectiveness in handling Parkinson's disease.

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Appendix A: Progress Report

Meeting Date	Summary of the Discussions	Achieved outcomes/ Tasks	Attendance
Mar 1 – Mar 7	• Research tremor frequency and EMG fundamentals • Study filtering & FFT needed • Perform basic IMU raw reading tests • Draft system block diagram	• Developed a clear understanding of tremor characteristics and EMG signal behavior • Identified the required signal processing techniques (filtering + FFT) • Successfully acquired initial IMU raw data • Established a preliminary system block diagram	Mohammed Ali Ahmed Ahmed
Mar 8 – Mar 14	• Finalize component selection • Assemble hardware (IMU + EMG) • Verify raw signal acquisition	• Finalized hardware component selection (IMU, EMG, microcontroller) • Successfully assembled the initial hardware setup • Verified proper acquisition of raw EMG and IMU signals	Mohammed Ali Ahmed Ahmed
Mar 15 – Mar 19	• Test IMU using generated frequencies • Compare measured vs real frequency • Begin basic filtering implementation • Identify signal errors	• Validated IMU frequency measurement using controlled test signals • Achieved close agreement between measured and actual frequencies • Implemented initial filtering techniques • Identified key signal noise and measurement errors	Mohammed Ali Ahmed Ahmed
Mar 24 – Mar 30	• Implement filtering (moving average, band-pass) • EMG rectification & RMS calculation • Begin real-time value output • Site visit to “Impact Physiotherapy Center”	• Implemented signal conditioning • Successfully applied EMG rectification and RMS envelope extraction • Achieved real-time signal visualization • Gained practical clinical insights from Impact Physiotherapy Center for the treatment phase	Mohammed Ali Ahmed Ahmed Dr. Mohammed
Mar 31 – Apr 6	• Extract tremor frequency & amplitude • Create simple tremor score logic • Compare rest vs motion signals	• Extracted tremor features • Developed an initial tremor scoring algorithm • Successfully distinguished between rest and movement signals	Mohammed Ali Ahmed Ahmed Dr. Ebrahim

	• The sensor burned down and made the case worse • Online meeting with the supervisor • Check the methods of treatment available internationally	• Identified hardware limitations after sensor failure and order another one after days due to the circumstances in the area • Explored existing treatment methods and clinical practices • Received technical feedback from supervisor	
Apr 7 – Apr 13	• Validate system response • Record and analyze data • Identify inconsistencies • Choose the best treatment method which is FES in our case and implementation • Do the analysis to verify whether it is applicable or not	• Validated system response under different conditions • Collected and analyzed experimental datasets • Identified inconsistencies in detection performance • Selected Functional Electrical Stimulation (FES) as the target treatment approach • Evaluated feasibility of integrating FES into the system	Mohammed Ali Ahmed Ahmed
Apr 14 – Apr 20	• Compare rest vs voluntary movement • Evaluate differentiation capability • Document findings • Deep research about the FES stimulation and its function and the logic behind it	• Successfully differentiated voluntary vs involuntary motion patterns • Evaluated system accuracy in tremor classification • Documented experimental findings • Developed deeper understanding of FES principles and control logic	Mohammed Ali Ahmed Ahmed
Apr 21 – Apr 27	• Improve filtering thresholds • Reduce noise/artifacts • Stabilize wiring and power • Finishing the paper about FES stimulation as a treatment	• Improved detection thresholds for higher accuracy • Reduced noise and motion artifacts significantly • Enhanced system stability • Completed technical write-up on FES-based treatment	Mohammed Ali Ahmed Ahmed

Apr 28 – May 7	• Perform repeated system trials • Begin long-duration stability test • Start to finalize prototype and 3D design • Prepare technical documentation	• Conducted repeated experimental trials to validate reliability • Initiated long-duration system stability testing • Finalized prototype structural design (3D model) • Prepared technical documentation	Mohammed Ali Ahmed Ahmed
May 8 – May 14	• Soldering the components in the stripboard • Prepare the final 3D printed case for the device • Put the final touch in the report to submit • Start preparing the presentation slides	• pelted hardware soldering on stripboard • assembled 3D-printed wearable enclosure • Completed and refined final project report • Prepared presentation slides for project demonstration	Mohammed Ali Ahmed Ahmed

Appendix B: Components Cost

TableB-1 Components Cost

	Component	Quantity	Price per Quantity (BD)
1	Arduino Nano 33 BLE Sense Rev2	1	28.783
2	EMG Sensor	2	7.92
3	ESP-12f NodeMcu D1 Mini	2	7.95
4	Batteries	2	3.000
5	3D Printed Models	2	7.5
6	Total	Total	55.153

Appendix C: Additional Figures

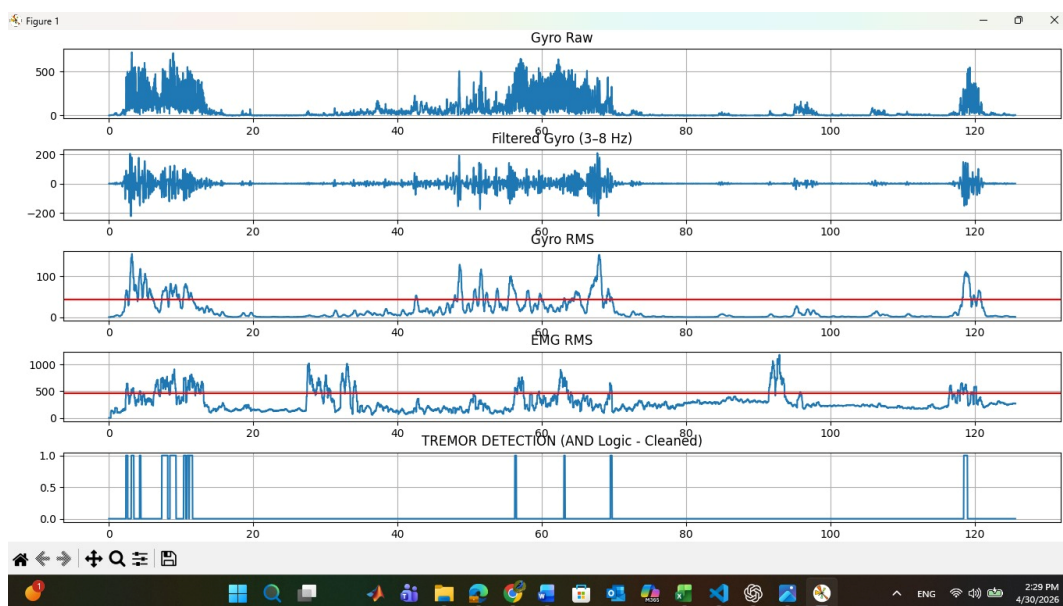


Figure (C-1) Open Source Data Analysis

1:11 PM Sun 29 Mar

EMG and Gyro...03-29 18-14-33

Home Insert Draw Page Layout Formulas

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	A	B	C	D	E	F	G	H	I	J
	myData0	myData1	myData2	myData3	myData4	myData5				
1	0.02433192	421.46701	0.20243071	217						
2	0.04240029	421.493988	0.18310545	217						
3	0.08868908	421.520996	0.13647875	215						
4	0.10193792	421.548004	0.12207031	217						
5	0.10193792	421.575012	0.2024307	216						
6	0.13960271	421.602997	0.18310545	216						
7	0.15972362	421.630005	0.18310545	221						
8	0.22264592	421.657013	0.19301011	208						
9	0.22275708	421.684998	0.13647875	214						
10	0.22286033	421.712006	0.18310545	217						
11	0.2643655	421.739014	0.20243071	216						
12	0.28047042	421.765991	0.18310545	216						
13	0.30974096	421.792999	0.25895023	218						
14	0.34018208	421.820007	0.13647875	220						
15	0.37249417	421.846985	0.10571599	215						
16	0.40216329	421.873993	0.14950499	219						
17	0.43738812	421.901001	0.17263348	220						
18	0.46617225	421.928009	0.18310545	218						
19	0.46629887	421.954987	0.22837263	213						
20	0.48965017	421.981995	0.12207031	211						
21	0.52021317	422.009003	0.20243071	215						
22	0.55390921	422.036011	0.13647875	216						
23	0.58578812	422.062988	0.19301011	220						
24	0.61070379	422.089996	0.14950499	218						
25	0.64220967	422.117004	0.13647875	215						
26	0.67024421	422.144012	0.13647875							
27	0.73076292									

mySet Metadata Device Metadata Time

Figure (C-2) Data Collected From Tre-Con

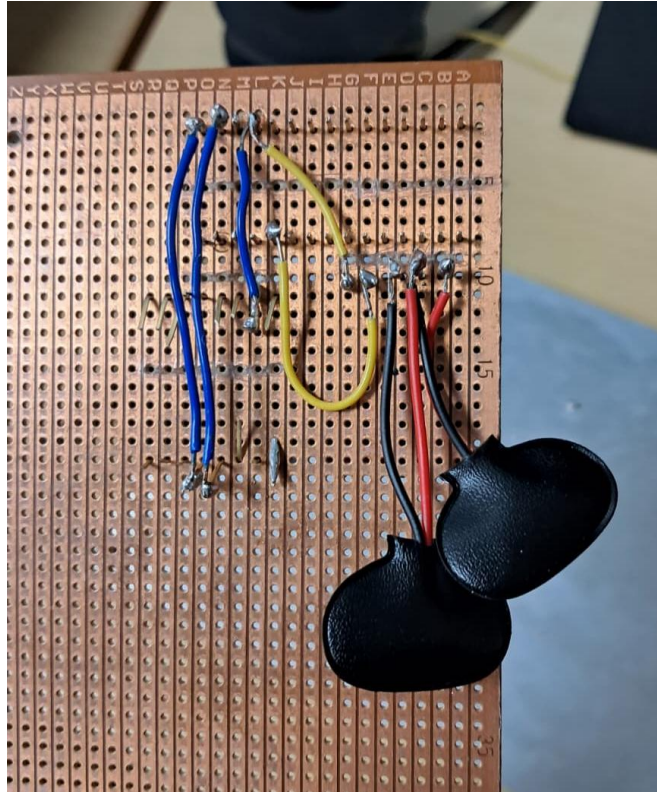
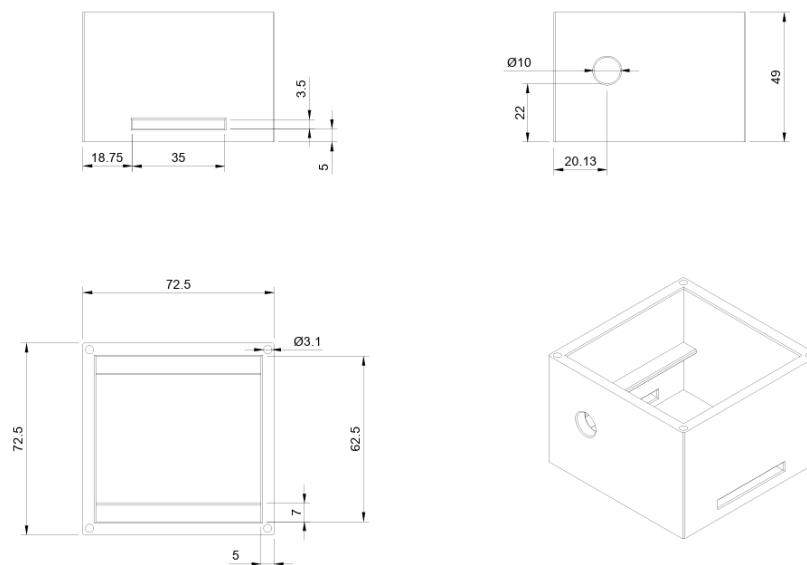


Figure (C-3) Circuit Soldering in a Stripboard



All measurements in mm

Figure (C-4) 3D Design Drawings

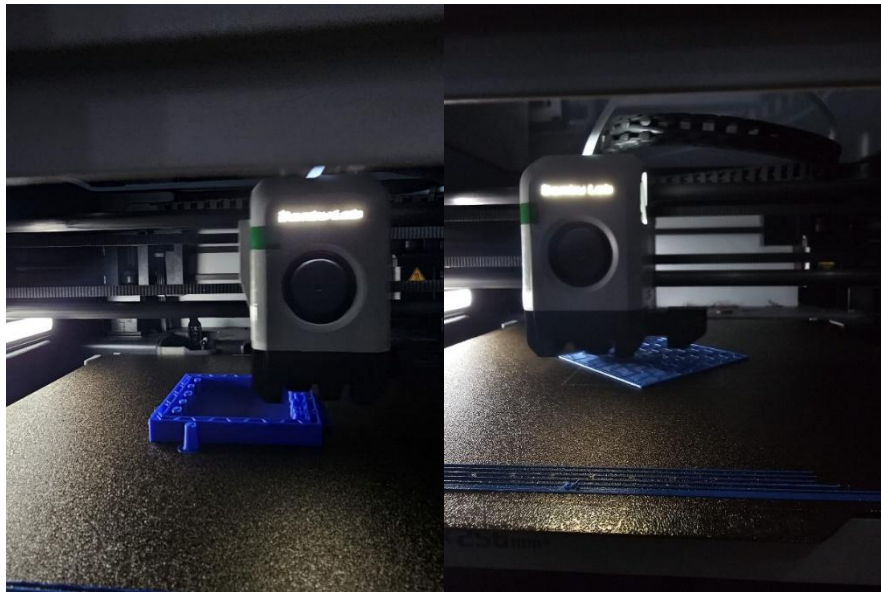


Figure (C-5) 3D Printing

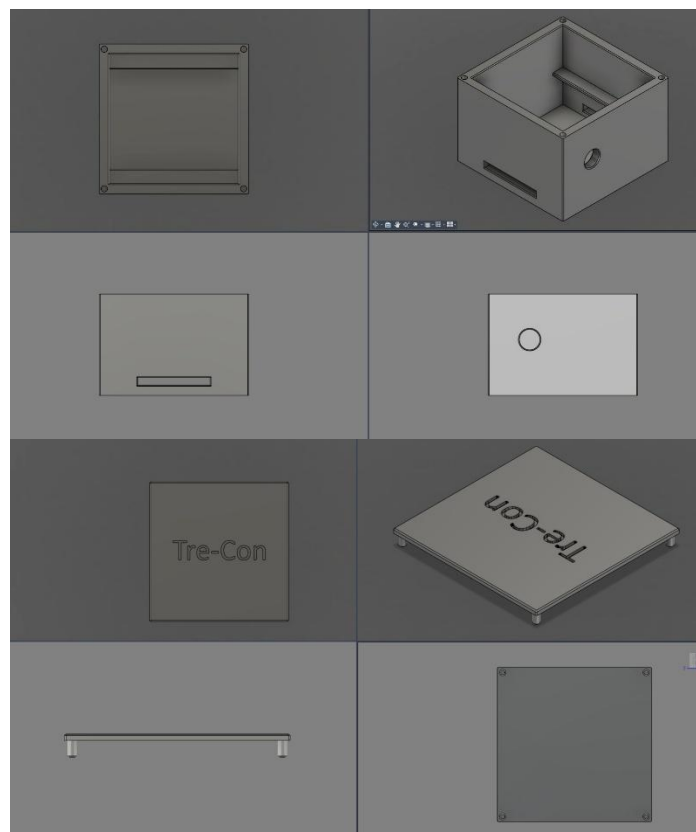


Figure (C-6) 3D Design

Appendix D: Coding references

The Python website code, Arduino nano 33 BLE REV2 code, Wi-Fi chip, and Blynk application all their coding has been generated and edited with the support of the AI assistance “Codex”. Below are the listed links for each code respectively:

website Python full code:

[https://drive.google.com/file/d/1i91tHO5BsdcrPHemGJOkFP9n_A0BW-Zx/view?usp=drive link](https://drive.google.com/file/d/1i91tHO5BsdcrPHemGJOkFP9n_A0BW-Zx/view?usp=drive_link)

Arduino nano full code:

[https://drive.google.com/file/d/1qS0v23TnP24q_cueBJFoVBedxMqUiv4T/view?usp=drive link](https://drive.google.com/file/d/1qS0v23TnP24q_cueBJFoVBedxMqUiv4T/view?usp=drive_link)

Wi-fi chip:

[https://drive.google.com/file/d/1RplqOilPo8Bt7U0oNyXFOOEJLztkLD7r/view?usp=drive link](https://drive.google.com/file/d/1RplqOilPo8Bt7U0oNyXFOOEJLztkLD7r/view?usp=drive_link)

Blynk Application Python Code:

[https://drive.google.com/file/d/1Aul9rwYdsxMeewQR6XbyWBkHJGySzaSd/view?usp=drive link](https://drive.google.com/file/d/1Aul9rwYdsxMeewQR6XbyWBkHJGySzaSd/view?usp=drive_link)

ملخص المشروع

تم تصميم هذا المشروع لإنشاء نظام ذكي وقابل للارتداء لمراقبة وتحليل اهتزاز اليد لدى المرضى الذين يعانون من اضطرابات الحركة، وخصوصاً مرض باركنسون. تكمن المشكلة الرئيسية في أن أساليب التشخيص التقليدية تعتمد عادةً على الفحص البصري داخل العيادات، مما قد لا يعكس الوضع الحقيقي للمريض في حياته اليومية. يقدم نظامنا حلاً لهذه المسألة من خلال تزويد وسيلة فعالة لمراقبة حالة المريض بشكل دائم ودقيق في محيطه الطبيعي.

(EMG) يعتمد النظام في أدائه على دمج نوعين من الحساسات الذكية؛ حيث يقوم حساس النشاط العضلي برصد الإشارات الكهربائية الناتجة عن العضلات، بينما يقوم حساس الحركة والجايروسكوب بقياس اهتزازات اليد واتجاهاتها بدقة. تُجمع هذه البيانات وتعالج بسرعة من خلال وحدة تحكم صغيرة، ثم تُرسل بشكل لاسلكي إلى قاعدة بيانات سحابية، مما يسهل تخزين المعلومات وإتاحتها للأطباء ومقدمي الرعاية لمتابعة الحالة عن بُعد. وفي أي وقت عبر المنصات الإلكترونية والتطبيقات الذكية.

كأداة رئيسية للتحليل؛ بسبب ما (Python) "فيما يخص معالجة البيانات وتصنيفها، تم استخدام لغة "بايثون" تقدمه من دقة عالية وميزات متنوعة ومتطورة في التعامل مع الإشارات الرقمية. تساعد هذه العملية في التخلص من التشويش وتمييز ترددات الرجفة المرضية بوضوح، ثم تصنيف شدتها إلى مستويات معينة، وعرض النتائج عبر واجهة تفاعلية تقدم رسوماً بيانية توضح تطور الحالة، مما يسهل على المختصين اتخاذ القرارات الطبية الصحيحة بناءً على تحليل دقيق وشامل. تم إنشاء غطاء خارجي قوي بواسطة الطباعة ثلاثية الأبعاد لحماية المكونات وضمان استقرار الجهاز أثناء التشغيل.

وفيما يتعلق بالجانب العلاجي، فقد أولى المشروع في هذه المرحلة اهتماماً بالجانب البحثي والنظري لتقنية كطريقة للحد من الرعشة. على الرغم من أهمية هذا القسم، إلا أنه لم يُنفذ (FES) التحفيز الكهربائي الوظيفي عملياً في النموذج الحالي بسبب عدة عوامل، منها قلة الوقت والضرورة لإجراءات طبية صارمة واختبارات دقيقة قبل التطبيق الفعلي. لذا، يركز العمل المقبل على الانتقال إلى مرحلة العلاج من خلال تنفيذ دائرة التحفيز الكهربائي، مع السعي لاختيار مكونات إلكترونية بديلة توفر دقة